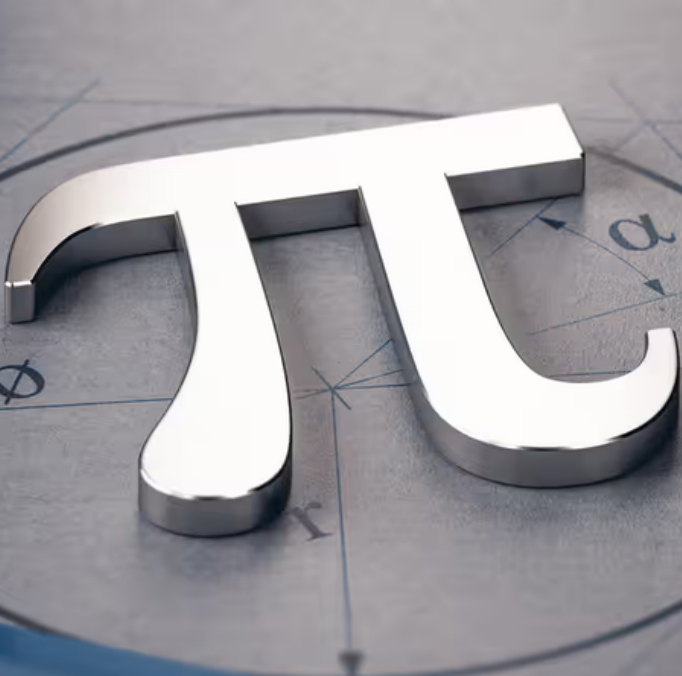




Topics covered

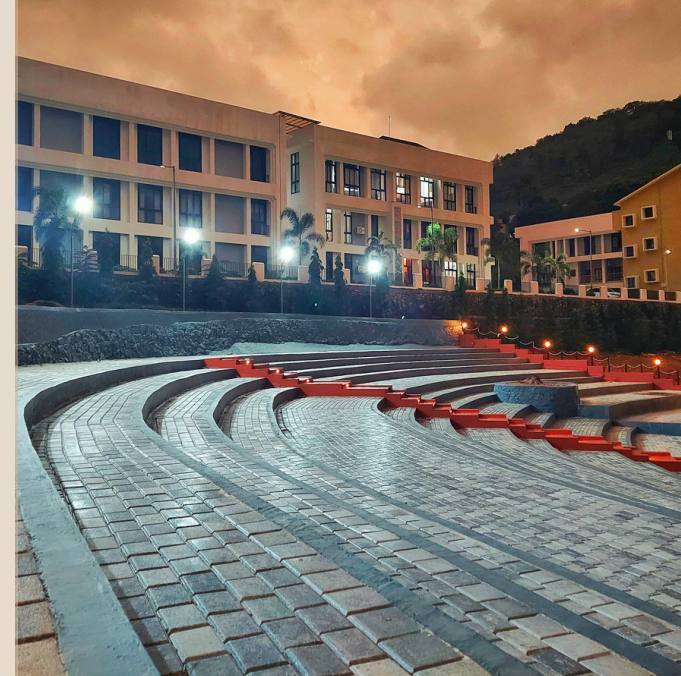
- Introduction of optimization and its significance in AI.
- Gradient descent algorithms: Strength, weakness, and applications.
- Regression and classification models: their optimization frameworks and solutions.
- Optimization algorithms for Deep learning models.
- Meta-heuristic algorithms: functionality and applications.
- ANN(Artificial Neural Network) based approximations for option pricing models.
- Machine Learning for solving fractional differential equations.
- Fuzzy sets, intuitionistic fuzzy sets, and their applications.
- Information systems and decision-making in a fuzzy environment.

Objectives of the workshop

- 1.To gain a deep understanding of optimization techniques and their critical role in solving real-world challenges.
- 2.To discover how differential equations play a pivotal role in shaping the behavior of ANN models.
- 3.To dive into the world of fuzzy computation for dealing with uncertainty and imprecision in data-driven decision-making.
- 4.To engage in practical exercises and coding sessions (using languages such as Python or Matlab)
- 5.To explore the interdisciplinary nature of mathematical computing and its intersections with AI\ML, data science, and engineering domains.
- 6.To understand the role of fuzzy logic behind every application in machine learning and artificial intelligence.

About the workshop

Join us for an immersive and enlightening one week workshop on Mathematical Computing, where we delve into three paramount topics: Optimization, Differential Equations for Artificial Neural Network (ANN) Models, and Fuzzy Computation. This workshop is tailored to equip participants with essential skills and knowledge at the intersection of mathematics and computing. Through theoretical insights and hands-on practical sessions, participants will embark on a transformative journey to harness the power of mathematical concepts for contemporary problem-solving.



Indian Institute of
Information Technology
Kottayam

One week workshop cum FDP on

MATHEMATICAL COMPUTING
(HYBRID MODE)

DECEMBER 11-17, 2023

**Department of Computational Science
and Humanities**

www.iiitkottayam.ac.in



ABOUT THE INSTITUTE

The Indian Institute of Information Technology, Kottayam is one among the IIITs that has been established as "Institutions of National Importance" by the Ministry of Education, Govt. of India under the ambit of IIIT (PPP) Act 2017. The institute offers 4 year B.Tech and B.Tech(Hons) program with specialized courses covering all major research domains of Computer Science and Electronics and Communication Engineering. The students are admitted against the JoSAA/CSAB counselling against the JEE main rank list. The institute also offers MTech in CSE and Ph.D. programs in CSE, ECE and Mathematics.

CONTACT

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Visit us at www.iiitkottayam.ac.in

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- ☎ Mr. Rohit Kumar - 9805654909
- ☎ Ms. Afila A S - 7736608056

ABOUT THE DEPARTMENT

The Department of Computational Science and Humanities was formally established in 2019 with an aim to develop mathematically sound young minds for solving industry-related problems arising from different industries and related areas. The department consists of nine full-time faculty members and 3 joint faculty members. It offers core courses in Mathematics/Statistics/Humanities to all B.Tech., M.Tech, and Ph.D. programs. This department takes pride in offering an exclusive doctoral program in Mathematics, meticulously crafted to meet the requirements of highly motivated and aspiring graduates holding a degree in this field.

RESOURCE PERSONS

- Dr. Kuldip Singh Patel, IIT Patna
- Dr. Jeevaraj, IIIT Gwalior
- Dr. Shiva Prasad Yadav, IIT Roorkee
- Dr. Pankaj Srivastava, NIT Allahabad
- Dr. Lakshmana Gomathi Nayagam, NIT Trichy
- Dr. Ritesh Kumar Dubey, SRM Chennai
- Dr. Jayakrushna Sahoo, IIIT Kottayam
- Dr. Susheel Kumar Joshi, IIIT Kottayam
- Dr. Manu Madhavan, IIIT Kottayam
- Dr. John Paul Martin, IIIT Kottayam
- Dr. P. Victor Paul, IIIT Kottayam
- Dr. Christina Terese Joseph, IIIT Kottayam
- Dr. Riyasudheen TK, IIIT Kottayam
- Ms. Suriyapriya K, IIIT Kottayam

PATRONS

- Prof. (Dr.) Rajiv Dharaskar, Director, IIIT Kottayam.
- Dr. M.Radhakrishnan, Registrar, IIIT Kottayam.
- Prof. (Dr.) P. Mohanan, Prof. In-Charge (Academics), IIIT Kottayam.

PROGRAM COORDINATORS

- Dr. T. Bakkyaraj, Assistant Professor and Head.
 - Dr. Susheel Kumar Joshi, Assistant Professor
 - Ms. Suriyapriya K, Assistant Professor
 - Dr. Riyasudheen TK, Assistant Professor
 - Dr. Dhanyamol M V, Assistant Professor.
- Department of Computational Science and Humanities, IIIT Kottayam*

FEE PAYMENT DETAILS

E-payment link: <https://tinyurl.com/bdhrvvf8>
Steps for payment
1. Choose the category as Educational Institutions
2. Choose payee as IIIT Kottayam
3. Choose payment category as Mathematical Computing workshop cum FDP

REGISTRATION

Target Audience: Faculty, PhD scholars and UG/PG students.

Registration deadline: 08/12/2023

Seats will be assigned on a first-come, first-served basis due to limited availability.)

Registration link: <https://tinyurl.com/2f4u3btj>

REGITRATION FEE

Participant type	Online	Offline*
PhD/PG/UG	1000/-	2000/-
Faculty	2000/-	3000/-

*Only working lunch and tea will be provided to the offline participants.

ACCOMODATION FEE DETAILS

Single room : ₹200/- per day
Shared room: ₹100/- per day
[Provided upon request via mail and mentioned the same in the registration link.]

ORGANISING TEAM

- Dr. Rubell Marion Lincy G
- Dr. Balu A
- Dr. Asha Sebastian
- Dr. Murugan D
- Dr. Gayathri G R
- Dr. Josit Mariya
- Dr. Mathew C D
- Mr. Anandhu Mohan
- Ms. Reetha Thomas
- Ms. Nicky Caroline S
- Ms. Julie Sara Benny
- Ms. Ansmol George
- Ms. Sheba Elizabeth Thomas
- Ms. Afila A S
- Ms. Fathima Nihla Latheef
- Ms. Mariya Thomas
- Mr. Rohit Kumar

